Problem: The Eagles tennis team played twenty matches last season, winning twelve matches and losing eight matches. They will play twenty matches again this year. What win and loss records are possible?

Solution plan:Build a table to show at least eight possible final season records.

Wins								
Losses								
Total	20	20						

Questions:

1. Write a formula that can be used to determine the number of losses L given the number of wins W.

   L = _____

2. Use your formula to determine the number of losses that correspond to the following numbers of wins:

   12 matches _____ 15 matches _____ 11 matches _____

3. Use the formula you wrote in question I to determine all the ways the Eagles can beat last year's record. Record your results in a table like the one used above.

4. Graph the data from the tables you constructed for this situation on a pair of coordinate axes and connect the points.

5. Does this problem represent a proportional relationship?_____ Support your answer using characteristics of both the formula and the graph of the data from this situation.

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